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Interfacial Engineering of Zn Anodes via Phase-Separated Conductive Polymers with Dual Ion-Electron Regulation

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ABSTRACT

Aqueous rechargeable zinc batteries (ARZBs) suffer from severe interfacial instability arising from nonuniform Zn^{2+} flux and parasitic electron-driven side reactions, which lead to dendrite growth, metal corrosion, and hydrogen gas evolution. Here, we employ poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate)/3-methylimidazolium:tetracyanoborate (PEDOT:PSS/*p*-MIM:TCB), a conductive polymer–ionic liquid composite that is stable in aqueous systems and forms effective Zn-ion channels through controlled phase separation, as an artificial protective layer. Furthermore, the PEDOT:PSS/*p*-MIM:TCB interfacial layer forms an electron-modulation layer owing to the work function difference with the Zn substrate, generating a uniform Zn^{2+} ion flux and effectively inhibiting Zn dendrite growth. As a result, Zn symmetric cells with PEDOT:PSS/*p*-MIM:TCB-coated electrodes exhibit stable cycling for 144 h at a high current density of 10 mA cm^{-2} with an areal capacity of 0.5 mAh cm^{-2}. Moreover, full cells paired with α-MnO_2 cathodes exhibit markedly enhanced rate capability, delivering a capacity advantage of approximately 100 mAh g^{-1} at a high current density of 1000 mA g^{-1}, indicating significantly improved power performance. This work presents an interfacial design strategy for stabilizing Zn metal anodes through concurrent ion-channel engineering and electronic structure modulation.

1. **Introduction**

Lithium-ion batteries (LIBs) are the most widely commercialized battery systems for electronics, electric vehicles, and smart grids because of their high energy density and efficiency.[1-3] However, scant reserves, limited ubiquity, and high cost of Li metals have driven the development of next-generation batteries, such as aluminum-ion,[4, 5] sodium-ion,[6-8] and Zn-ion[9-11] batteries, as alternatives to LIBs. Among them, aqueous rechargeable zinc batteries (ARZBs) are particularly promising for large-scale energy storage systems owing to the low cost of Zn, high theoretical capacity (820 mAh g^{-1}), and excellent rate performance.[12, 13] Moreover, the flammable organic electrolytes of LIBs can cause the explosion of batteries; however, ARZBs operate in aqueous electrolytes, resulting in additional safety benefits.[14, 15]

Despite the recent advances in the field of high-performance cathodes and electrolytes, the instability of the anode–electrolyte interface remains a critical challenge for the application of ARZBs, limiting their stability and cyclability.[16-20] In particular, the widely used Zn metal anodes are limited by interfacial issues, such as dendrite formation, corrosion, and hydrogen gas evolution. Furthermore, nonuniform Zn deposition and subsequent dendrite growth pose considerable difficulties.[21, 22] Owing to the high Young's modulus of Zn (108 GPa), Zn dendrites can easily penetrate the separator, increasing the risk of internal short circuits and severely reducing the battery lifespan.[23, 24]

To address the issues related to Zn dendrite growth, various strategies, such as the introduction of artificial protective layers (APLs),[25-27] electrolyte modifications,[28, 29] and electrode design,[30, 31] have been extensively investigated. Among them, the incorporation of APLs is widely adopted because of their simple fabrication and substantial effectiveness. APLs alleviate the Zn dendrite growth by regulating the electric field or ion diffusion and physical shielding. However, conventional APLs have predominantly been introduced with a focus on ion homogenization or electron regulation.

For ion homogenization in APLs, numerous coating materials comprising selective pores or ion channels have been explored.[32, 33] Polymer-based materials, renowned for their strong adhesion to substrates facilitated by abundant functional groups, are widely used as APLs. The inherent polar functional groups of polymer-based

APLs facilitate ion transport by forming ion channels via the regulation of Zn^{2+} kinetics.[34, 35] Based on Sand's time model, the realization of a uniform electric field distribution induces uniform Zn deposition, regulating Zn nucleation and growth to inhibit dendrite formation.[36, 37] Carbon-based materials with a high electrical conductivity, including carbon nanotubes,[38] carbon black,[39] and reduced graphene oxide,[40] have been employed as coating materials and facilitate surface charge transfer as electron regulators. However, because of their large surface area and high electrical conductivity, these materials tend to easily aggregate, which disrupts the uniform distribution of the electric field.[41] Another strategy for generating a uniform electric field is the manipulation of the electrochemical reaction at the electrode–electrolyte interface by adjusting the work function.[42] The work function difference between the coating materials and the electrode induces electron transfer, creating a negatively charged region that ensures a uniform electric field and prevents dendrite formation.[43] However, the simultaneous realization of uniform electron regulation and ion homogenization in APL design remains challenging.

Here, we introduce a conductive polymer film for fabricating dendrite-free Zn anodes using poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) and 3-methylimidazolium:tetracyanoborate (*p*-MIM:TCB), which are capable of uniformly regulating both the electrons and Zn^{2+} ions. PEDOT:PSS is attracting considerable attention in the fields of bioelectronics and energy storage devices because of its high conductivity, flexibility, cost-effectiveness, and easy processing.[44, 45] Particularly, PEDOT:PSS is widely utilized in battery cathodes,[46-49] anodes,[50-53] and binders[54-58] owing to its excellent electrochemical and thermal stability. However, its water solubility is prone to cause structural instability when applied to ARZBs operating in aqueous systems. To address this challenge, we propose the use of a mixed interfacial layer based on PEDOT:PSS and the *protic* ionic liquid (*p*-IL), *p*-MIM:TCB, which enhances the water resistance of PEDOT:PSS by effectively removing excess PSS, ensuring the stability of PEDOT:PSS/*p*-MIM:TCB layers in aqueous systems.[59] Moreover, the difference in the work function between the PEDOT:PSS/*p*-MIM:TCB interfacial layer and the Zn metal results in a uniform Zn^{2+}-ion flux, suppressing the Zn dendrite formation. Therefore, the PEDOT:PSS/*p*-MIM:TCB@Zn symmetric cell demonstrates highly stable Zn plating/stripping at 10 mA cm^{-2} and 0.5 mAh cm^{-2}.

Furthermore, the PEDOT:PSS/*p*-MIM:TCB@Zn|α-MnO_2 full cell exhibits considerably improved cycling and rate performances compared with the bare Zn|α-MnO_2 full cell. This study emphasizes the crucial effects of the simultaneous control of ion pathways and electric fields in the APL on metal anodes.

2. Results and discussion

Fig. 1 illustrates the Zn^{2+}-ion migration within the PEDOT:PSS/*p*-MIM:TCB coating layer. Owing to the ion exchange between PEDOT:PSS and *p*-MIM:TCB, the PEDOT:PSS layer separates into PEDOT-rich and PSS-rich domains.[59] The PSS-rich domains, which are enriched with sulfonic groups, exhibit a high adsorption affinity toward Zn^{2+} ions, forming Zn^{2+}-ion transport channels.[60, 61] Considering that these transport channels in the anode are critical determinants of Zn^{2+}-ion diffusion kinetics, the formation of PSS-rich domains via controlled phase separation in the PEDOT:PSS/*p*-MIM:TCB film is expected to facilitate the rapid movement of Zn^{2+} ions, enabling efficient plating and stripping at high current densities. Furthermore, compared with the Zn anode, the high work function PEDOT:PSS/*p*-MIM:TCB film effectively modulates the electron transfer at the interface by establishing an ohmic contact. This characteristic promotes a uniform electric field distribution and a well-regulated Zn^{2+}-ion flux, contributing to the suppression of Zn dendrite formation.

To further characterize the formation and structural features of the PEDOT:PSS/*p*-MIM:TCB protective layer, the morphology and chemical composition of the coated Zn electrode were investigated, as shown in **Fig. 2**. As schematically illustrated in **Fig. 2**a, ion exchange between PEDOT:PSS and *p*-MIM:TCB weakens the electrostatic interactions between $PEDOT^+$ and PSS^-, thereby promoting phase separation into PEDOT-rich and PSS-rich domains. The top-view SEM image in **Fig. 2**b shows that the PEDOT:PSS/*p*-MIM:TCB layer uniformly covers the Zn surface, while the cross-sectional SEM image in **Fig. 2**c confirms the formation of a continuous coating layer with a thickness of approximately 34.65 μm. The chemical structures of PEDOT:PSS and *p*-MIM:TCB are presented in **Fig. 2**d. Furthermore, SEM–EDS mapping and the corresponding elemental distributions reveal the homogeneous distribution of C, N, O, and S throughout the coating layer (**Fig. 2**e and **2**f), supporting the uniform incorporation of the coating components over the Zn surface. The FT-IR spectrum of PEDOT:PSS/*p*-

MIM:TCB@Zn exhibits characteristic vibrational bands associated with the coating components (**Fig. 2**g), further confirming the successful formation of the PEDOT:PSS/*p*-MIM:TCB layer on the Zn substrate.

The *p*-MIM:TCB-induced solubility change is confirmed by comparing the ultraviolet–visible–near-infrared (UV–Vis–NIR) absorption spectra of bare PEDOT:PSS and PEDOT:PSS/*p*-MIM:TCB thin films before and after rinsing with deionized water (DI-W) (**Fig. 3**a and **3**b). The bare PEDOT:PSS thin films exhibit no noticeable absorption spectrum after DI-W rinsing owing to dissolution. Conversely, the PEDOT:PSS/*p*-MIM:TCB thin films remain unchanged, exhibiting no spectral variations before and after rinsing with DI-W. These observations suggest that the ion exchange between PEDOT:PSS and *p*-MIM:TCB coupled with excess PSS removal forms a water-resistant film. Furthermore, the UV-Vis spectra of the DI-W collected after rinsing show negligible absorbance compared with the characteristic absorption of PSS and *p*-MIM:TCB, indicating minimal release of these components during rinsing (**Fig. S2**). The absorbance difference before and after rinsing further decreases with increasing *p*-MIM:TCB content and becomes nearly negligible in the presence of *p*-MIM:TCB, defining a water-resistant region (**Fig. S1**). These results suggest that the ion exchange between PEDOT:PSS and *p*-MIM:TCB, coupled with excess PSS removal, promotes the formation of a water-resistant film.

Fig. 3c presents a scanning electron microscopy (SEM) image of the PEDOT:PSS/*p*-MIM:TCB film after immersion in DI-W, confirming its phase-separated morphology. The incorporation of *p*-MIM:TCB induces phase separation within the PEDOT:PSS matrix, leading to the formation of interconnected PEDOT-rich domains and PSS-rich domains. Subsequent rinsing with DI water selectively removes excess PSS from the film, leaving voids between the PEDOT-rich domains and thereby developing a three-dimensional porous morphology. The resulting interconnected structure increases the accessible surface area and provides pathways for ion transport within the film.[59] This porous structure accommodates the volume expansion and improves the electrolyte wettability. Additionally, the Zn^{2+} affinity of PSS^{-} ions in the PSS-rich domain promotes a uniform Zn^{2+}-ion flux (**Fig. 1**), leading to dendrite-free Zn deposition.

The linear current-voltage (I-V) characteristics of the PEDOT:PSS/*p*-MIM:TCB-coated Zn electrode demonstrate a nonrectifying ohmic contact between the Zn substrate and the coating layer (**Fig. 3**d). Linear I-V meas-

urements were performed using a probe station (**Fig. S3**) The PEDOT:PSS/*p*-MIM:TCB-coated Zn anode exhibits linear *I–V* characteristics, demonstrating a nonrectifying characteristic, indicating an ohmic contact between the Zn substrate and the coating layer.[43, 62] The ohmic contact facilitates electron transfer across the Zn/PEODT:PSS/*p*-MIM:TCB interface, while the higher work function of the coating layer relative to Zn, as discussed below, drives electron redistribution near the interface. This interfacial electron redistribution results in an accumulated negative charge in the coating layer, which can attract cations and promote their inward diffusion, thereby contributing to more uniform Zn deposition.[43] These findings are consistent with the electron diffusion model predicted based on the work functions of the Zn metal (−3.74 eV) and PEDOT:PSS (−5.00 eV).[64, 65]

To further validate the experimentally observed Zn^{2+} modulation by the PEDOT:PSS/*p*-MIM:TCB coating layer, density functional theory (DFT) and coarse-grained molecular dynamic simulations were performed to examine the electronic and ionic transport properties. The enhanced dendrite suppression in PEDOT:PSS/*p*-MIM:TCB@Zn was hypothesized to originate from the work function difference between the PEDOT:PSS/*p*-MIM:TCB film and Zn metal. Simulations confirmed that the pristine Zn (0001) surface exhibits a relatively low work function of 3.80 eV (**Fig. 3**e). In contrast, when PEDOT, PSS, *p*-MIM, and TCB monomers were placed on the Zn surface, the work function of PEDOT: PSS/*p*-MIM:TCB@Zn interface model increased to 4.99 eV (**Fig. 3**f), thereby inducing a strong electronic driving force for interfacial charge redistribution. The work function of each model was determined by independently aligning its planar-averaged electrostatic potential to its own vacuum level (0 eV); accordingly, the value obtained for the interface model represents the effective work function of the coated Zn surface rather than a modified work function of bulk Zn. Notably, the oscillatory potential profile within the Zn slab remains essentially unchanged upon adsorption, indicating that the work-function increase originates from the interfacial dipole formed by the adsorbed layer, across which the potential variation is localized—consistent with the general behavior of metal–semiconductor contacts. This increase in work function is consistent with the experimentally observed ohmic contact (**Fig. 3**e) and accounts for the formation of a negatively charged interfacial region that attracts Zn^{2+} ions, promoting homogeneous plating. The close

correlation between the calculated work function and the observed uniform Zn morphology highlights the pivotal role of interfacial electronic modulation in suppressing dendrite growth.

The electrolyte wettability of the Zn surface was further evaluated by contact angle measurements. As shown in **Fig. S6**, the electrolyte contact angle decreased from 95.3° for bare Zn to 76.0° for PEDOT:PSS/*p*-MIM:TCB@Zn, indicating substantially improved electrolyte wettability after coating. The enhanced wettability promotes more intimate contact between the electrolyte and the electrode surface, facilitating Zn^{2+} transport across the electrode/electrolyte interface and thereby contributing to more uniform Zn deposition.

The interfacial stability of the Zn electrode was further examined by linear sweep voltammetry (LSV) and Tafel polarization measurements. As shown in **Fig. S7**a, the PEDOT:PSS/*p*-MIM:TCB@Zn electrode exhibits suppressed hydrogen evolution behavior compared with bare Zn, indicating that the coating layer mitigates water-related parasitic reactions at the Zn/electrolyte interface. Consistently, the Tafel polarization curves in **Fig. S7**b show that the corrosion current decreased from 27.7 μA for bare Zn to 17.6 μA for PEDOT:PSS/*p*-MIM:TCB@Zn, while the corrosion potential shifted slightly from −0.015 to −0.012 V vs. Zn/Zn^{2+}. The reduced corrosion current indicates improved corrosion resistance of the coated Zn electrode, further confirming the protective effect of the PEDOT:PSS/*p*-MIM:TCB layer against interfacial side reactions. The chemical stability of the coating in the electrolyte was also evaluated by immersing PEDOT:PSS/*p*-MIM:TCB@Zn in 3 M $Zn(OTf)_2$ electrolyte for 10 h. As shown in **Fig. S8**, the coating retains its surface morphology after immersion, indicating sufficient structural stability under the aqueous electrolyte environment. In addition, the composition of the coating layer was optimized by EIS measurements using Zn symmetric cells containing different *p*-MIM:TCB contents ranging from 40 to 190 wt% relative to PEDOT:PSS (**Fig. S9**). Based on the interfacial resistance obtained from the EIS measurements, a *p*-MIM:TCB content of 130 wt% relative to PEDOT:PSS was selected as the optimized composition. The effect of coating thickness was further evaluated using a thicker PEDOT:PSS/*p*-MIM:TCB coating of approximately 50 μm (**Fig. S10**). The thicker coating exhibited increased interfacial

resistance compared with the optimized coating, indicating that excessive coating thickness hinders efficient Zn^{2+} transport. Accordingly, the coating thickness was controlled to approximately 34.65 μm to balance interfacial protection with efficient ion transport.

The effects of the PEDOT:PSS/*p*-MIM:TCB film on the cycling stability of the Zn anode were investigated using symmetric Zn|Zn cells at a high current density of 10 mA cm^{-2} and an areal capacity of 0.5 mAh cm^{-2}. As illustrated in **Fig. 4**a, a short circuit occurs in the bare Zn symmetric cell after 44 h of cycling, suggesting that the grown Zn dendrites have perforated the separator. Conversely, PEDOT:PSS/*p*-MIM:TCB@Zn maintains a stable charge–discharge behavior for approximately 144 h. Enlarged voltage profiles near the failure points are presented in **Fig. S11**, further highlighting the markedly prolonged cycling stability of the coated Zn electrode compared with bare Zn. The improved stability of the symmetric cells indicates that the inhibition of Zn dendrite growth is induced by the phase-separated PEDOT:PSS/*p*-MIM:TCB film possessing a porous structure.

The cycling stability was further evaluated under a lower current density of 1 mA cm^{-2} and an areal capacity of 1 mAh cm^{-2} (**Fig. S12**). Under these conditions, the bare Zn symmetric cell failed after approximately 80 h, whereas the PEDOT:PSS/*p*-MIM:TCB@Zn cell maintained stable cycling for approximately 360 h, demonstrating that the protective effect of the coating is retained across different plating/stripping conditions. To further assess the durability of the Zn anode under a practically relevant depth of discharge (DOD), galvanostatic cycling was conducted at 10% DOD with an areal capacity of 5.85 mAh cm^{-2} (**Fig. S13**). The PEDOT:PSS/*p*-MIM:TCB@Zn symmetric cell exhibited more stable and prolonged cycling behavior than the bare Zn cell, further demonstrating the effectiveness of the coating in stabilizing Zn plating/stripping under increased Zn utilization. The rate capability of Zn anodes was evaluated by cycling symmetric cells at various current densities, ranging from 1 to 20 mA cm^{-2} at a constant areal capacity of 2 mAh cm^{-2}. From **Fig. 4**b, a short-circuit phenomenon occurs in bare Zn during cycling at a rate of 5 mA cm^{-2}. However, PEDOT:PSS/*p*-MIM:TCB@Zn displays substantially more stable plating/stripping voltage profiles than bare Zn during cycling. This result can be attributed to the high electrical conductivity of the PEDOT:PSS/*p*-MIM:TCB layer(**Fig.S5**), which is capable of regulating the electric field that becomes progressively nonuniform with increasing current density. Moreover, the electrochemical impedance spec-

troscopy (EIS) measurements of the symmetric cells (**Fig. 4**c) demonstrate that the interfacial resistance decreases with the introduction of the PEDOT:PSS/*p*-MIM:TCB film. These findings indicate that rapid Zn^{2+}-ion transport and electronic induction are achieved by PEDOT:PSS/*p*-MIM:TCB-based mixed ionic-electronic conductors.

To further investigate the nucleation behavior, interfacial electrochemical behavior, and reversibility of Zn plating/stripping, Zn|Cu asymmetric cells were assembled. **Fig. 4**d and **4**e present the nucleation overpotentials measured at current densities of 1 and 5 mA cm^{-2}, respectively. PEDOT:PSS/*p*-MIM:TCB@Cu exhibited lower nucleation overpotentials than bare Cu at both current densities, with a low value of 34 mV at 1 mA cm^{-2}. The reduced nucleation overpotential indicates a lower energy barrier for the initial nucleation of Zn on the Cu substrate, suggesting facilitated Zn nucleation at the modified Cu interface.

The reversibility of Zn plating/stripping was further evaluated by Coulombic efficiency (CE) measurements. Specifically, Zn with an areal capacity of 1 mAh cm^{-2} was plated onto Cu foil at a current density of 2 mA cm^{-2} and subsequently stripped at the same current density until the voltage reached 0.5 V. As shown in **Fig. 4**f, the Zn|PEDOT:PSS/*p*-MIM:TCB@Cu asymmetric cell exhibited relatively stable Zn plating/stripping behavior with an average CE of 98.3% over 100 cycles. In contrast, the Zn|bare Cu asymmetric cell showed pronounced fluctuations in CE during cycling.

The corresponding voltage profiles of the Zn|bare Cu and Zn|PEDOT:PSS/*p*-MIM:TCB@Cu asymmetric cells are compared in **Fig. 4**g and **4**h. The Zn|PEDOT:PSS/*p*-MIM:TCB@Cu cell exhibited a low polarization potential of 91 mV, substantially lower than that of the Zn|bare Cu cell (409 mV). Because Zn deposition in this asymmetric-cell configuration occurs on a Cu-based substrate, the measured polarization reflects the interfacial behavior associated with Zn plating on Cu, which differs from that observed in Zn|Zn symmetric cells. Collectively, the reduced nucleation overpotential and polarization potential, together with the more stable CE behavior, indicate that the PEDOT:PSS/*p*-MIM:TCB layer promotes favorable Zn deposition behavior on the Cu substrate, improves Zn plating/stripping reversibility, and facilitates interfacial electrochemical processes.

The PEDOT:PSS/*p*-MIM:TCB film and the surface and cross section of the electrode before and after cycling were investigated via field emission (FE)-SEM (**Fig. 5**). **Fig. 5**a and 5b reveal clear differences between the bare

Zn and PEDOT:PSS/*p*-MIM:TCB@Zn metal anodes. From **Fig. 5**a, numerous Zn^{2+} ions are deposited on bare Zn to form dendrites. **Fig. 5**b demonstrates the SEM image of the Zn surface after peeling off the PEDOT:PSS/*p*-MIM:TCB coating layer following 400 cycles, which confirms uniform Zn deposition at the interface because of the consistent Zn^{2+}-ion flux formed by the PEDOT:PSS/*p*-MIM:TCB film. The thickness of the Zn metal anode was measured using SEM after 500 cycles at a current density of 10 mA cm^{-2} and an areal capacity of 0.5 mAh cm^{-2} to determine whether Zn deposition occurred uniformly or not. **Fig. 5**c and **5**d display the cross-sectional SEM images of bare Zn before and after cycling, respectively, and the thickness of the Zn plate increases from 251 to 345 μm during cycling. Conversely, the thickness of PEDOT:PSS/*p*-MIM:TCB@Zn remains unchanged during cycling (**Fig. 5**e and **5**f), indicating that the coating layer induces a more uniform deposition of Zn^{2+} ions and a reversible plating/stripping behavior. The structural stability of the PEDOT:PSS/*p*-MIM:TCB coating during repeated Zn plating/stripping was further examined by SEM. As shown in **Fig. S14**, the coated Zn electrode retained its surface morphology after both 1 and 150 cycles at 10 mA cm^{-2} and 0.5 mAh cm^{-2}, indicating that the coating layer remains structurally stable during prolonged cycling. Furthermore, the Zn deposition behavior was directly compared using top-view and cross-sectional SEM images after plating capacities of 1 and 2 mAh cm^{-2} at 2 mA cm^{-2} (**Fig. S15**). In contrast to the increasingly nonuniform Zn deposition observed on bare Zn, PEDOT:PSS/*p*-MIM:TCB@Zn exhibited a more uniform and compact deposition morphology, demonstrating that the coating layer effectively regulates Zn^{2+} flux and promotes homogeneous Zn growth.

Full cells were fabricated using an α-MnO_2 cathode to investigate the practical applicability of the PEDOT:PSS/*p*-MIM:TCB@Zn anode. **Fig. S16** presents the X-ray diffraction (XRD) pattern of hydrothermally synthesized α-MnO_2, exhibiting peaks indexed to tetragonal α-MnO_2 with the space group I4/m.[66] The lattice parameters are calculated as a = b = 9.78 Å and c = 2.86 Å.[66] The crystal structure of α-MnO_2, which exhibits a 2 × 2 tunnel configuration, is presented in **Fig. S17**. **Fig. 6**a illustrates the cyclic voltammetry (CV) profiles at a scan rate of 0.5 mV s^{-1}. Distinctive redox peaks are observed for the α-MnO_2 cathodes in both the bare Zn and PEDOT:PSS/*p*-MIM:TCB@Zn cells, which is consistent with the prior reports.[67, 68] This result confirms that despite the coating of the Zn metal with the PEDOT:PSS/*p*-MIM:TCB layer, Zn^{2+} ions reversibly move between

the anode and cathode. Furthermore, **Fig. 6**b depicts the voltage profiles for the first charge–discharge cycle of the bare Zn|α-MnO_2 and PEDOT:PSS/*p*-MIM:TCB@Zn|α-MnO_2 cells at 400 mA g^{-1}, demonstrating similar voltage plateaus. Additionally, the PEDOT:PSS/*p*-MIM:TCB@Zn|α-MnO_2 cell exhibits a higher discharge capacity (231.9 mAh g^{-1}) than the bare Zn|α-MnO_2 cell (156 mAh g^{-1}) at a rate of 400 mA g^{-1}. The cycling performance (**Fig. 6**c) reveals that the PEDOT:PSS/*p*-MIM:TCB@Zn|α-MnO_2 cell exhibits higher discharge capacities than the bare Zn|α-MnO_2 cell. Specifically, the PEDOT:PSS/p-MIM:TCB@Zn|α-MnO_2 cell delivered an initial discharge capacity of 231.9 mAh g^{-1} and maintained 160 mAh g^{-1} after 100 cycles, whereas the bare Zn|α-MnO_2 cell showed a lower initial discharge capacity of 156 mAh g^{-1}, which decreased to 112 mAh g^{-1} after 100 cycles. **Fig. 6**d presents the rate performances of the bare Zn|α-MnO_2 and PEDOT:PSS/*p*-MIM:TCB@Zn|α-MnO_2 cells at various current densities, ranging from 100 to 1000 mA g^{-1}. Additionally, **Fig. S18** and **S19** display the corresponding voltage profiles under these conditions. The average discharge capacities of the PEDOT:PSS/*p*-MIM:TCB@Zn|α-MnO_2 cells at the current densities of 100, 200, 500, and 1000 mA g^{-1} are 242, 232, 201, and 176 mAh g^{-1}, respectively, which are higher than those of bare Zn|α-MnO_2 cells. The overall full-cell performance reaffirms the substantial role of the PEDOT:PSS/*p*-MIM:TCB coating layer in improving the rate capability. To further evaluate the practical applicability of the PEDOT:PSS/*p*-MIM:TCB coating under increased Zn utilization, full-cell cycling tests were performed using thinner Zn foils with a thickness of 100 μm. As shown in **Fig. S20**, the PEDOT:PSS/*p*-MIM:TCB@Zn|α-MnO_2 full cell exhibited more stable cycling performance than the bare Zn|α-MnO_2 cell, demonstrating that the beneficial effect of the coating is retained even when a thinner Zn anode is employed.

Zn^{2+} density profiles near the polymer interfaces further substantiate the dominant role of PSS in Zn^{2+} ion transport. The equilibrated simulation cell (**Fig. 7**a) clearly reveals phase-separated PEDOT-rich and PSS-rich domains. Notably, the Zn^{2+} density distribution profile indicates a pronounced preference for Zn^{2+} ions to reside within the PSS-rich regions, with a local concentration substantially higher than in the PEDOT-rich domains. This analysis reinforces the conclusion that PSS-rich domains serve as primary Zn^{2+} transport channel, effectively guiding ion flux at the electrode surface.

Fig. 7b presents the radial distribution functions (RDFs) for Zn^{2+} ions with respect to PEDOT and PSS chains. The sharp and intense peak in the Zn^{2+}–PSS RDF adjacent to $r \approx 0.43$ nm indicates a strong affinity between Zn^{2+} ions and the sulfonate groups of PSS. In contrast, Zn^{2+}–PEDOT RDF exhibits a broader and significantly weaker first peak, reflecting comparatively weaker interactions. In absolute terms, this association is moderate rather than trapping: the potential of mean force obtained directly from the RDF, $F(r) = -k_B T \ln g(r)$,[69] exhibits a first-shell well depth of only ~0.7 $k_B T$ (~1.8 kJ mol^{-1} at 300 K) for the Zn^{2+}–PSS pair, with a barrier of only ~0.8 $k_B T$ for escape from the first coordination shell, indicating that Zn^{2+}–sulfonate association and dissociation are thermally facile. Consistently, trajectory analysis (**Fig. 7**c) shows that Zn^{2+} ions migrate through successive coordination with sulfonate groups along the PSS chains, with exchanges between neighboring sulfonate sites occurring on the nanosecond timescale (i.e., finite residence times at individual sites rather than permanent trapping), highlighting the pathway-forming nature of ion transport within the PSS-rich domains. Notably, the Zn^{2+} ions remain strongly hydrated even in the immediate vicinity of the PSS chains: the Zn^{2+}–water RDFs computed for all Zn^{2+} ions and for the subset of Zn^{2+} ions located within 1.5 nm of the PSS chains both exhibit pronounced first peaks that are considerably higher than the first peak of the Zn^{2+}–PSS RDF (**Fig. S21**). This indicates that Zn^{2+} coordinates with the sulfonate groups without complete desolvation, such that hydrated Zn^{2+} ions are readily accommodated and transported within the water-swollen PSS-rich channels.

To quantitatively assess this phase-separated transport mechanism, the diffusion coefficients of Zn^{2+} ions within different domains of PEDOT:PSS were evaluated (**Fig. 7**d). Zn^{2+} ions within a cutoff distance of 0.43 nm from either PEDOT or PSS chains were selected based on the RDF analysis. Zn^{2+} ions exhibited higher diffusivity in the vicinity of PSS chains compared to PEDOT chains. Specifically, the median diffusion coefficient of Zn^{2+} ions near PSS chains is 4.85×10^{-7} cm^2 s^{-1}, whereas that near PEDOT chains is lower at 3.94×10^{-7} cm^2 s^{-1}. While this difference in mobility is moderate (~23%), ionic conductivity is proportional to the product of ion concentration and mobility. Given the substantially higher local Zn^{2+} concentration in the PSS-rich region (**Fig. 7**a), the combined effect of higher local ion density and higher diffusivity indicates a considerably greater effective Zn^{2+} conductivity within the PSS-rich domains, further confirming their role as a Zn^{2+} ion channel within the phase-

separated structure of the interlayer. We note that these diffusion coefficients were evaluated for Zn^{2+} ions located within 0.43 nm of the respective chains, that is, for ions in direct contact with the coordinating groups; the higher mobility observed near the PSS chains therefore directly excludes a trapping scenario, in which strong sulfonate coordination would reduce rather than enhance Zn^{2+} mobility. The Zn^{2+}–sulfonate interaction thus lies in a moderate-binding regime that is beneficial for channel-mediated transport—strong enough to preferentially confine Zn^{2+} within the PSS-rich channels (**Fig. 7a**), yet weak enough to permit facile site-to-site exchange (**Fig. 7**c) without imposing a significant desolvation or migration barrier (**Fig. S21**).

Overall, these simulation results establish that PSS-rich regions serve as preferential pathways for Zn^{2+} ions transport, enabling directional and rapid ion migration across the interlayer. This atomistic transport behavior is consistent with the experimentally observed uniform Zn deposition and prolonged cycling stability, and it provides a mechanistic basis for the dendrite-suppressing effect of the PEDOT:PSS/*p*-MIM:TCB coating, as reflected in the improved cycling stability (**Fig. 4**a) and reduced interfacial resistance (**Fig. 4**c).

3. Conclusions

In summary, we have demonstrated the suppression of Zn dendrite growth by coating the Zn metal using PEDOT:PSS/*p*-MIM:TCB layers for application in ARZBs. The work function difference between the Zn anode and the PEDOT:PSS/*p*-MIM:TCB coating layer establishes a uniform electric field distribution at the interface, effectively regulating local electron flux. Concurrently, the phase-separated morphology of PEDOT:PSS induced by the *p*-MIM:TCB additive provides efficient Zn^{2+}-ion transport pathways, facilitating rapid ion diffusion. By simultaneously realizing uniform electron conduction and accelerated Zn^{2+} transport, this dual-regulation coating layer synergistically suppresses dendritic Zn growth and significantly enhances Zn plating/stripping reversibility. Consequently, the PEDOT:PSS/*p*-MIM:TCB-coated Zn anodes exhibit stable and long cycling durability for 144 h even at a high current density of 10 mA cm^{-2}. Furthermore, the PEDOT:PSS/*p*-MIM:TCB@Zn|α-MnO_2 cell demonstrates considerably improved rate capability, cycling performance, and high capacity. Atomistic modeling further confirms the work function modulation imposed by the coating materials and indicates that Zn^{2+} ions pref-

erentially migrate through the PSS-rich network rather than the PEDOT network. The lifetime of the present system is limited primarily by the mechanical adhesion between the polymer film and the Zn substrate rather than by dendrite growth, which represents a major limitation of the current design. Improving this interfacial adhesion will therefore be a key focus of our future work. The phase-separated conductive polymer film possessing a high work function and its porous structure provide durable and efficient coating for rechargeable metal batteries, ensuring uniform metal-ion deposition/stripping through electron-accumulation layers and ion-transport channels even at high current densities.

CRediT authorship contribution statement

S.L., J.L., M.K., and J. S. contributed equally to this work. K.W.N., B.H.L., and H.P. supervised the project. M.K., E.N., and K.K. characterized the conductive polymer. J.L. and S.H.L. synthe-sized the conductive polymer. J. S. conducted DFT and MD simulations. S.L., J.L., M.K., J.S., H.K., E.N., and J.S. co-wrote the manuscript. K.W.N., B.H.L., and H.P. revised the manuscript. All authors discussed the results and assisted during the manu-script preparation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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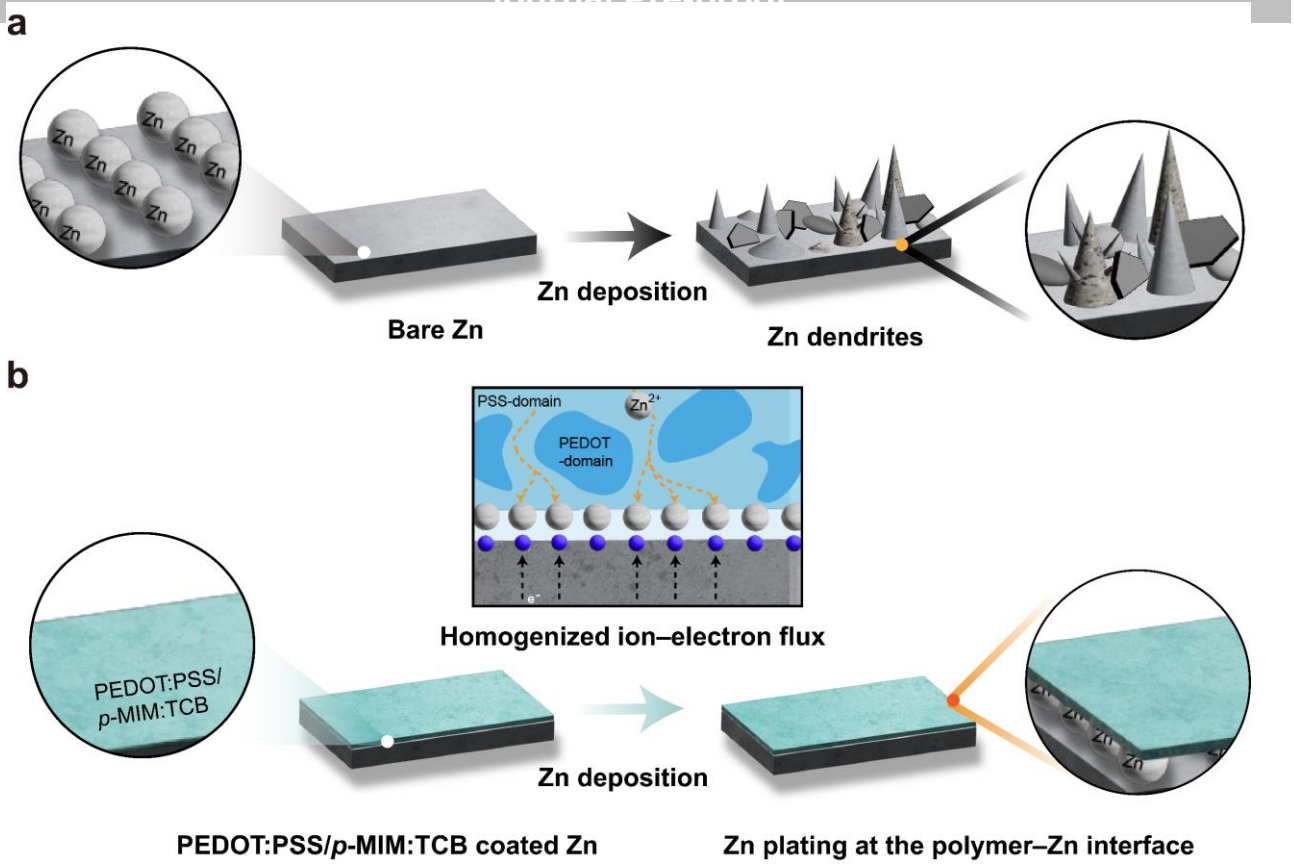


Fig. 1. Schematic illustration of Zn deposition behavior on bare and PEDOT:PSS/*p*-MIM:TCB-coated Zn anodes. (a) Non-uniform Zn deposition on the bare Zn anode, accompanied by dendrite formation and parasitic interfacial reactions. (b) Proposed interfacial regulation by the PEDOT:PSS/*p*-MIM:TCB coating, which facilitates uniform Zn deposition and suppresses dendrite growth and parasitic interfacial reactions.

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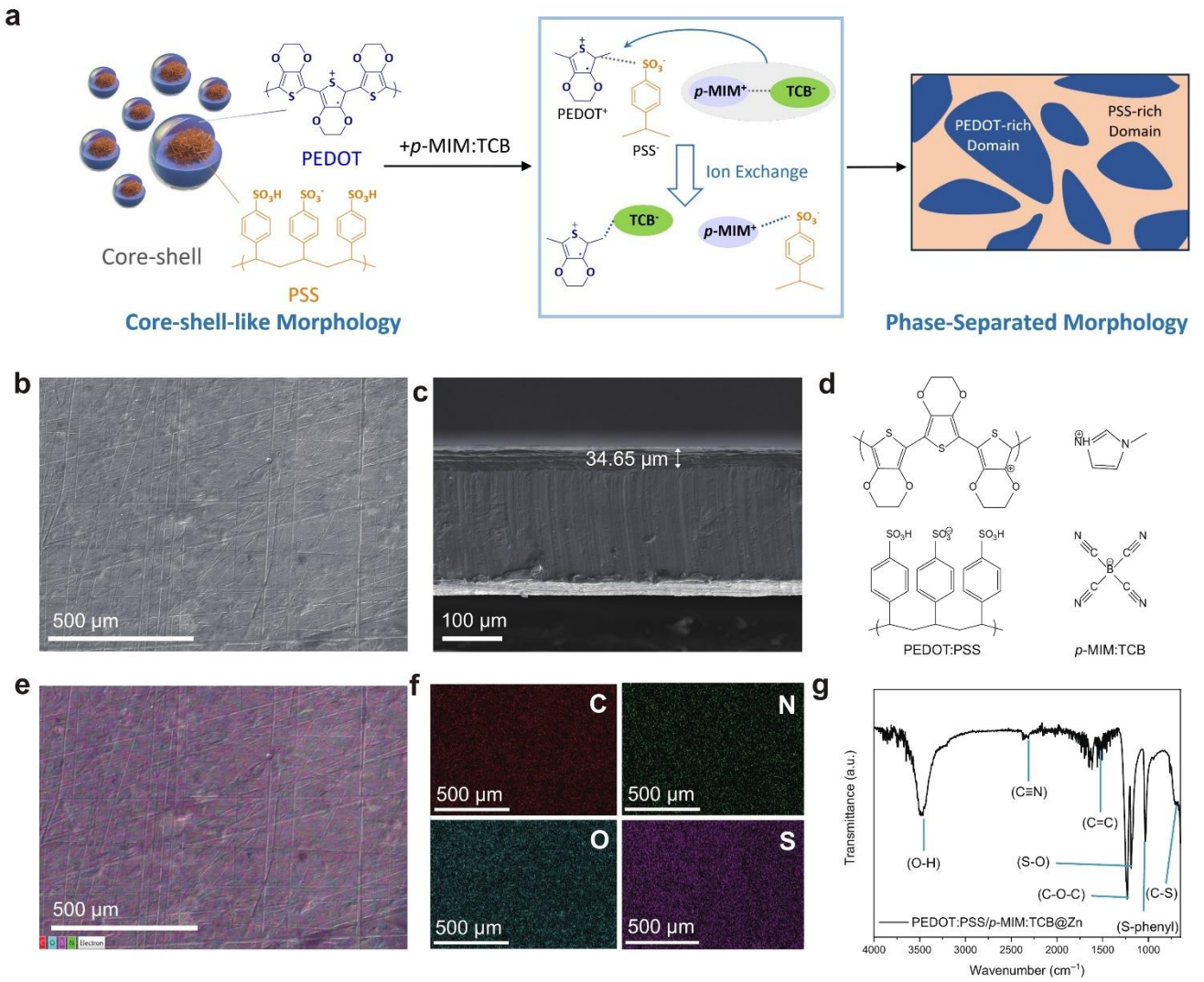


Fig. 2. (a) Schematic illustration of the interaction between PEDOT:PSS and *p*-MIM:TCB and the resulting phase separation induced by ion exchange. (b) Top-view SEM image and (c) cross-sectional SEM image of PEDOT:PSS/*p*-MIM:TCB@Zn, showing a coating thickness of approximately 34.65 μm. (d) Chemical structures of PEDOT:PSS and *p*-MIM:TCB. (e) SEM–EDS mapping image and (f) corresponding elemental distributions of C, N, O, and S in the PEDOT:PSS/*p*-MIM:TCB coating. (g) FTIR spectrum of PEDOT:PSS/*p*-MIM:TCB@Zn, showing the characteristic functional groups of the coating components.

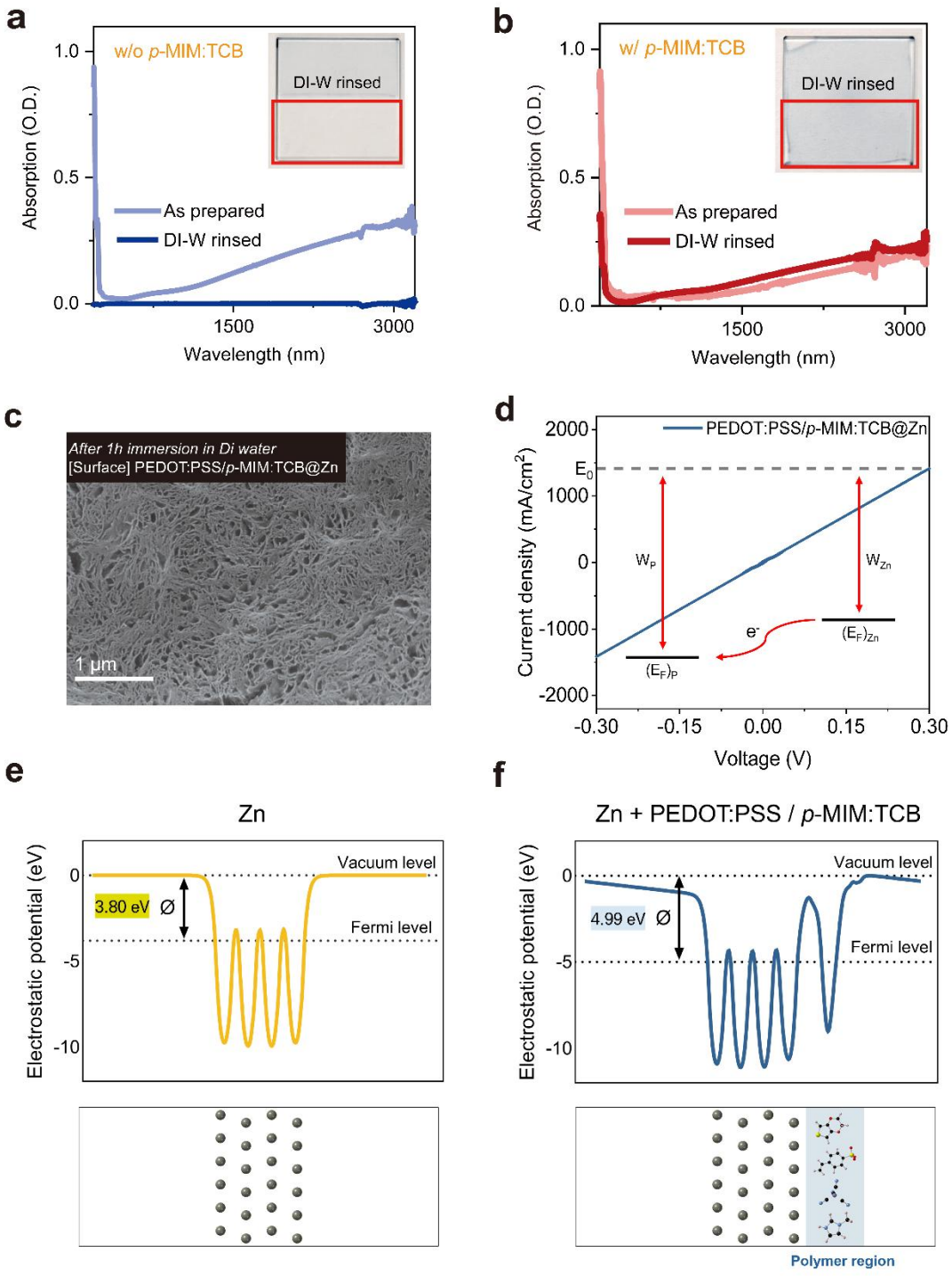


Fig. 3. (a) PEDOT:PSS and (b) PEDOT:PSS/*p*-MIM:TCB thin films before and after rinsing with DI-W. Insets depict the images of the thin films, with half of each thin film immersed in DI-W. (c) Surface SEM image of the PEDOT:PSS/*p*-MIM:TCB-coated Zn anode after immersion in DI-W for 1 h. (d) I–V characteristics of the Zn anode modified by the PEDOT:PSS/*p*-MIM:TCB coating layer. The planar-averaged electrostatic potential profile along the surface-normal axis of (e) the bare Zn slab and (f) the PEDOT:PSS/*p*-MIM:TCB@Zn interface model. The vacuum level of each slab model is independently aligned to 0 eV.

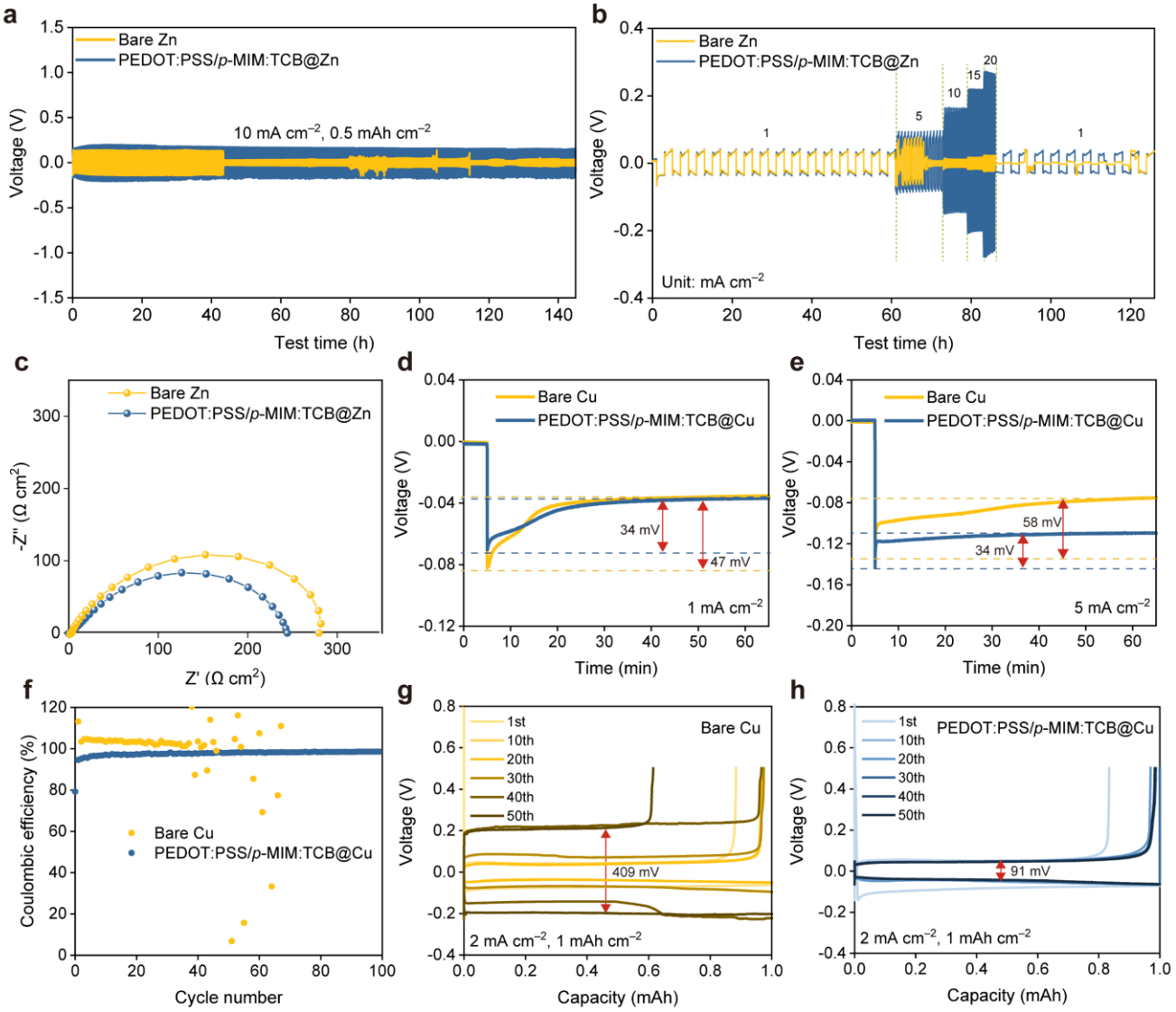


Fig. 4. Electrochemical test results of the Zn|Zn symmetric cells and Zn|Cu asymmetric cells. (a) Galvanostatic charge–discharge profiles of the symmetric Zn cells comprising bare Zn and PEDOT:PSS/*p*-MIM:TCB@Zn at a current density of 10 mA cm^{-2} and an areal capacity of 0.5 mAh cm^{-2}. (b) Rate performance at the current densities of 1–20 mA cm^{-2}. (c) Nyquist plots of the symmetric cells composed of bare Zn and PEDOT:PSS/*p*-MIM:TCB@Zn. Nucleation overpotential of bare Cu and PEDOT:PSS/*p*-MIM:TCB@Cu asymmetric cells at the current densities of (d) 1 and (e) 5 mA cm^{-2}. (f) CE determined for 100 cycles at a current density of 2 mA cm^{-2} and an areal capacity of 1 mAh cm^{-2} and the corresponding charge–discharge voltage profiles of (g) bare Cu and (h) PEDOT:PSS/*p*-MIM:TCB@Cu.

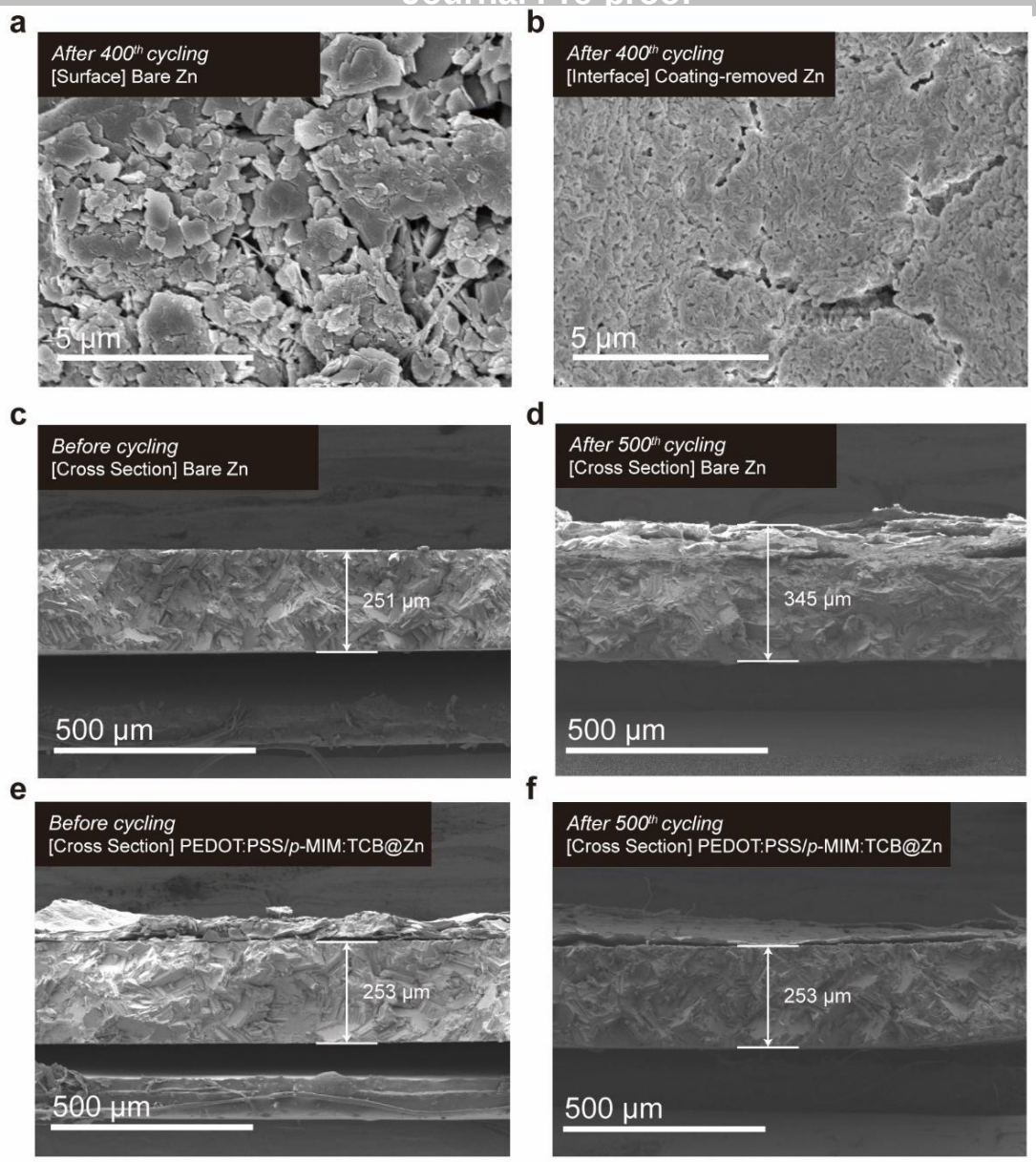


Fig. 5. Morphological evolution of Zn electrodes during prolonged cycling. Surface SEM images of (a) bare Zn and (b) coating-removed PEDOT:PSS/*p*-MIM:TCB@Zn after 400 cycles. Cross-sectional SEM images of Zn deposition on (c, d) bare Zn and (e, f) PEDOT:PSS/*p*-MIM:TCB@Zn after 500 cycles. Plating/stripping conditions are 10 mA cm^{-2} and 0.5 mAh cm^{-2}.

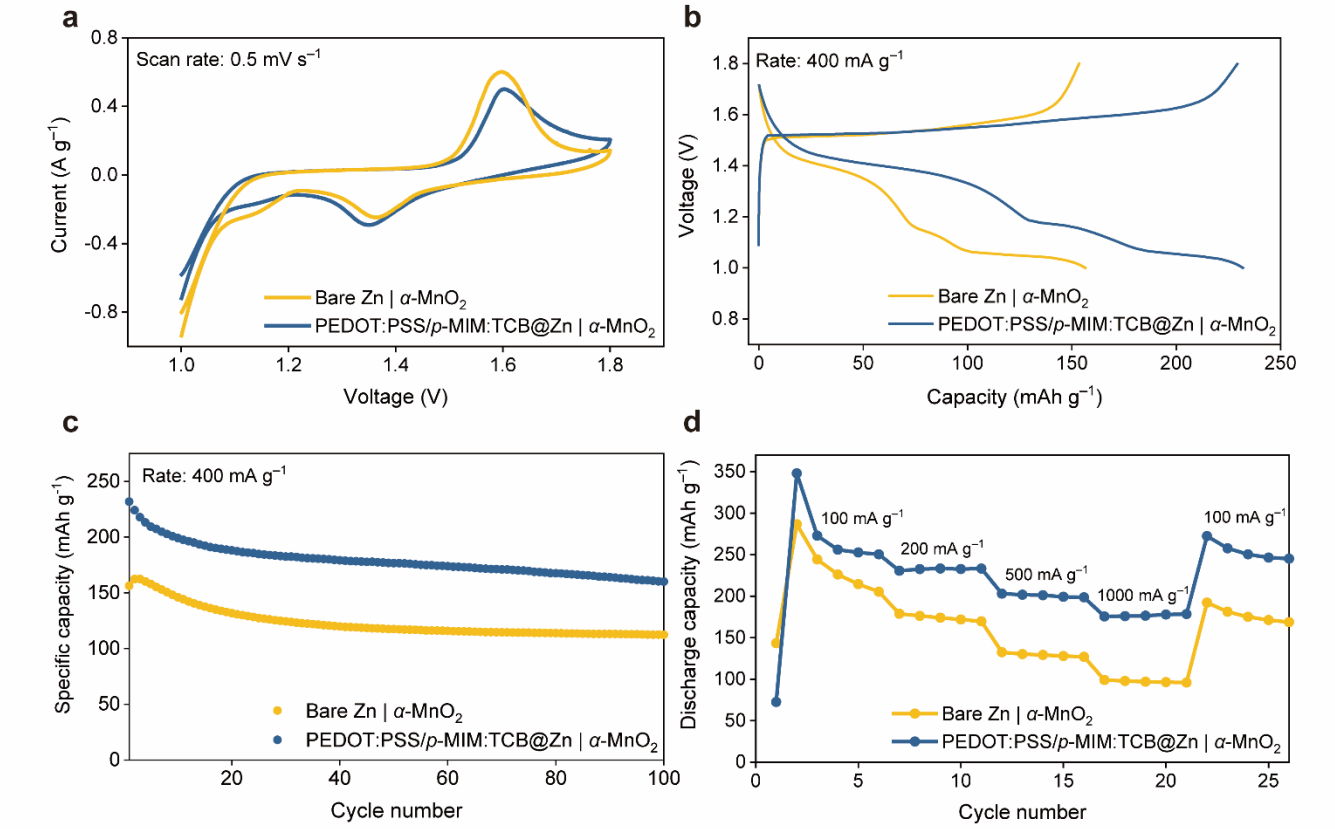


Fig. 6. Electrochemical performance of Zn|α-MnO_2 full cells. (a) CV curves measured at 0.5 mV s^{-1} and (b) charge–discharge voltage profiles measured at 400 mA g^{-1} for bare Zn|α-MnO_2 and PEDOT:PSS/*p*-MIM:TCB@Zn|α-MnO_2 full cells. (c) Cycling performances and (d) rate capabilities of bare Zn|α-MnO_2 and PEDOT:PSS/*p*-MIM:TCB@Zn|α-MnO_2 full cells.

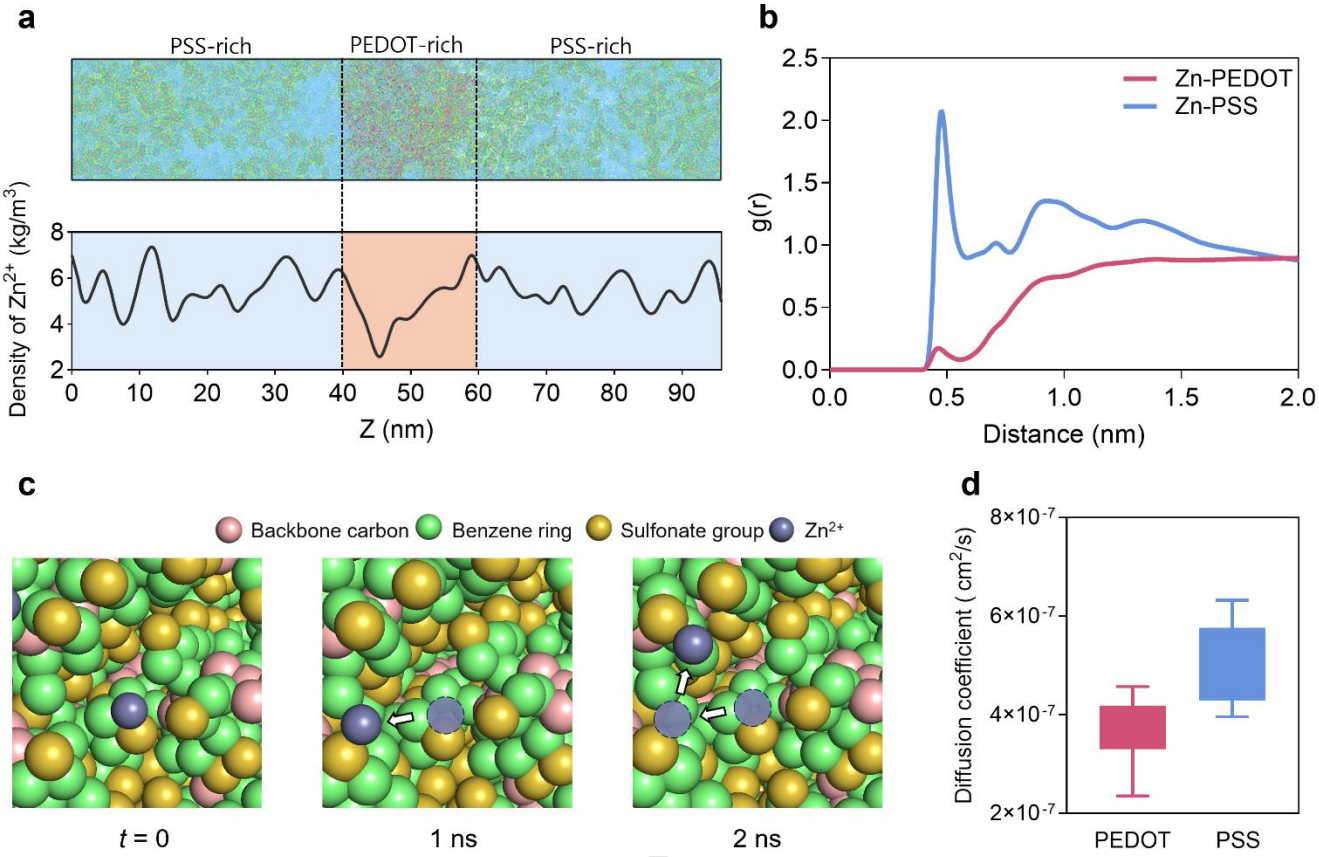
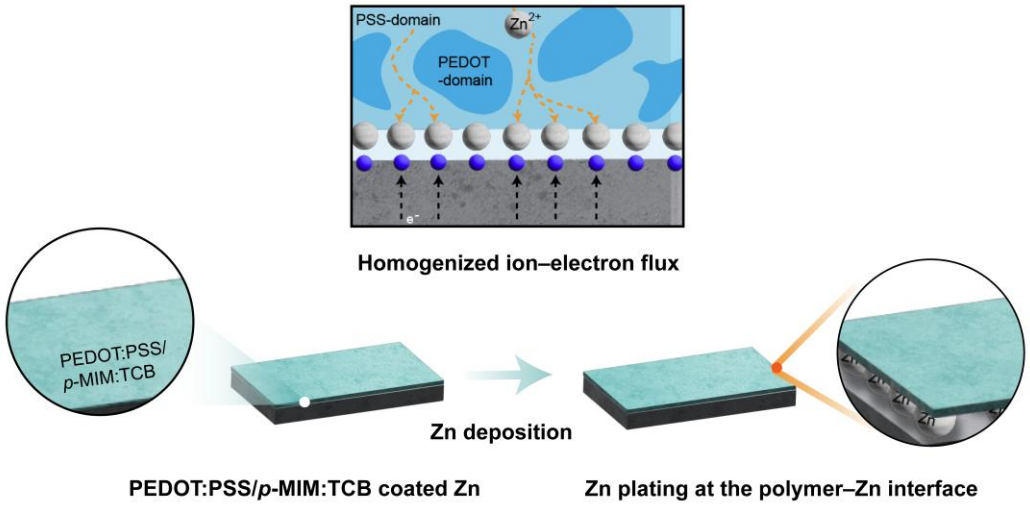


Fig. 7. Coarse-grained molecular dynamics simulations of Zn-ion transport within the phase-separated PEDOT:PSS. (a) Simulation cell of PEDOT:PSS and the density profiles of Zn^{2+} ions in PEDOT:PSS. (b) RDFs g(r) for Zn^{2+} ions with respect to PEDOT and PSS, respectively. (c) Visualization for Zn^{2+} ions transport through PSS chain. (d) Boxplots of Zn^{2+} ions diffusion coefficient near PEDOT and PSS chain.



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Highlights

- PEDOT:PSS/p-MIM:TCB composite protects Zn anodes in aqueous batteries.
- Phase separation in the composite creates effective Zn-ion transport channels.
- Electron-modulation layer ensures uniform Zn2+ flux and inhibits dendrites.
- Protected Zn enables stable cycling at 10 mA cm-2 and enhanced rate capability.

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